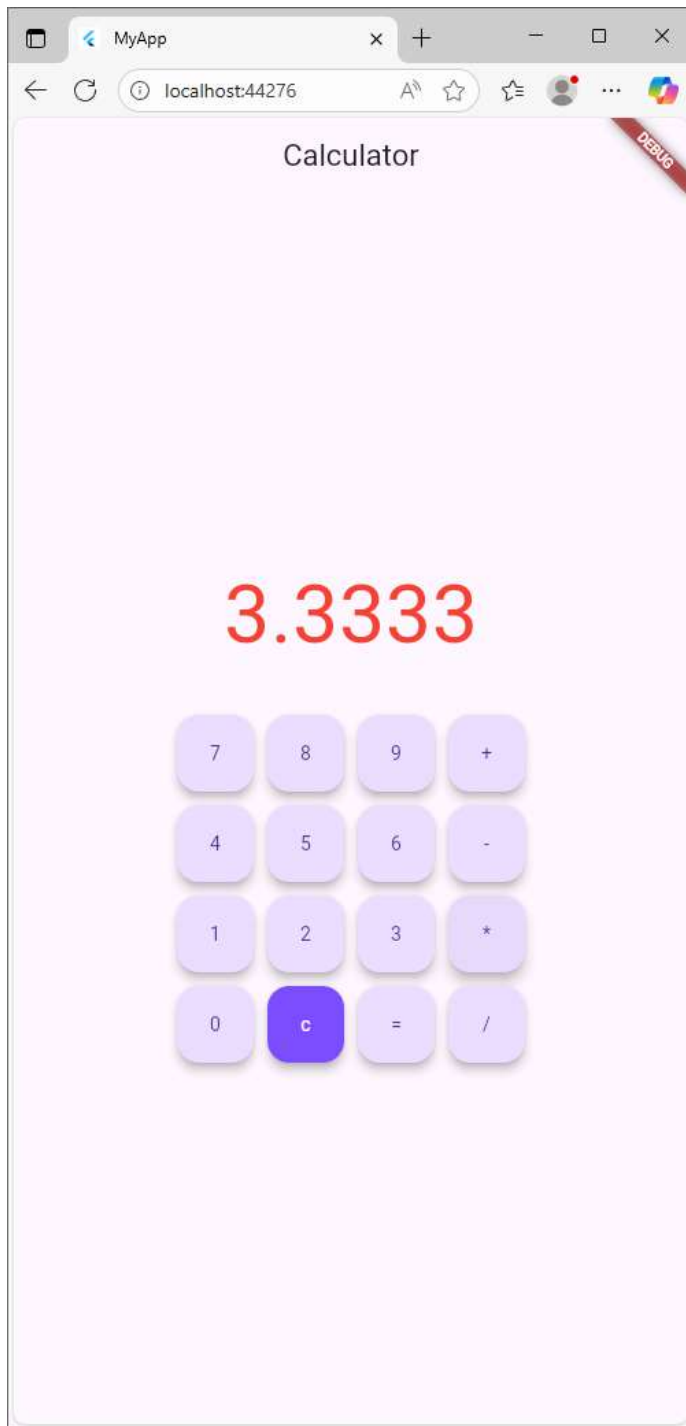




1

## CW2 Basic Calculator

2

```

1 import 'package:flutter/material.dart';
2
3 void main() {
4   runApp(const MyApp());
5 }
6
7 class MyApp extends StatelessWidget {
8   const MyApp({super.key});
9
10  @override
11  Widget build(BuildContext context) {
12    return const MaterialApp(title: "New Calculator", home: MyHomePage());
13  }
14 }
15
16 class MyHomePage extends StatefulWidget {
17   const MyHomePage({super.key});
18
19   @override
20   State<MyHomePage> createState() => _MyHomePageState();
21 }
22
23 class _MyHomePageState extends State<MyHomePage> {
24   String num1 = "";
25   String num2 = "";
26   String operator = "";
27   String display = "0";
28
29   Widget calcBtn(String text, VoidCallback onPressed) {
30     return Padding(
31       padding: const EdgeInsets.symmetric(horizontal: 6.0),
32       child: FloatingActionButton(
33         onPressed: onPressed,
34         child: Text(text, style: TextStyle(fontSize: 24)),
35       ),
36     );
37   }
38
39   void pressNumber(String value) {
40     setState(() {
41       if (display == "0" && operator.isEmpty) {
42         num1 = value;
43         display = num1;
44       } else if (operator.isEmpty) {
45         num1 += value;
46         display = num1;
47       } else if (num1.isNotEmpty && operator.isNotEmpty && display == "0") {
48         num2 = value;
49         display = num2;
50       } else {
51         num2 += value;
52         display = num2;
53       }
54     });
55   }
56
57   void pressOperator(String op) {
58     setState(() {

```

```

59     operator = op;
60   });
61 }
62
63 void clear() {
64   setState(() {
65     num1 = "";
66     num2 = "";
67     operator = "";
68     display = "0";
69   });
70 }
71
72 void calculate() {
73   setState(() {
74     double n1 = double.parse(num1);
75     double n2 = double.parse(num2);
76     double result = 0;
77
78     switch (operator) {
79       case "+":
80         result = n1 + n2;
81         break;
82       case "-":
83         result = n1 - n2;
84         break;
85       case "*":
86         result = n1 * n2;
87         break;
88       case "/":
89         result = n1 / n2;
90         break;
91     }
92
93     display = double.parse(result.toStringAsFixed(4)).toString();
94     num1 = result.toString();
95     num2 = "";
96     operator = "";
97   });
98 }
99
100 @override
101 Widget build(BuildContext context) {
102   return Scaffold(
103     appBar: AppBar(
104       title: Center(child: Text("NEW CAL")),
105       backgroundColor: Colors.purple[200],
106     ),
107     body: Center(
108       child: Column(
109         mainAxisAlignment: .center,
110         children: [
111           Text(display, style: TextStyle(fontSize: 40, color: Colors.red)),
112           SizedBox(height: 30),
113           Row(
114             mainAxisAlignment: .center,
115             children: [
116               calcBtn("7", () => pressNumber("7")),
117               calcBtn("8", () => pressNumber("8")),
118               calcBtn("9", () => pressNumber("9")),
119               calcBtn("+", () => pressOperator("+")),
120

```

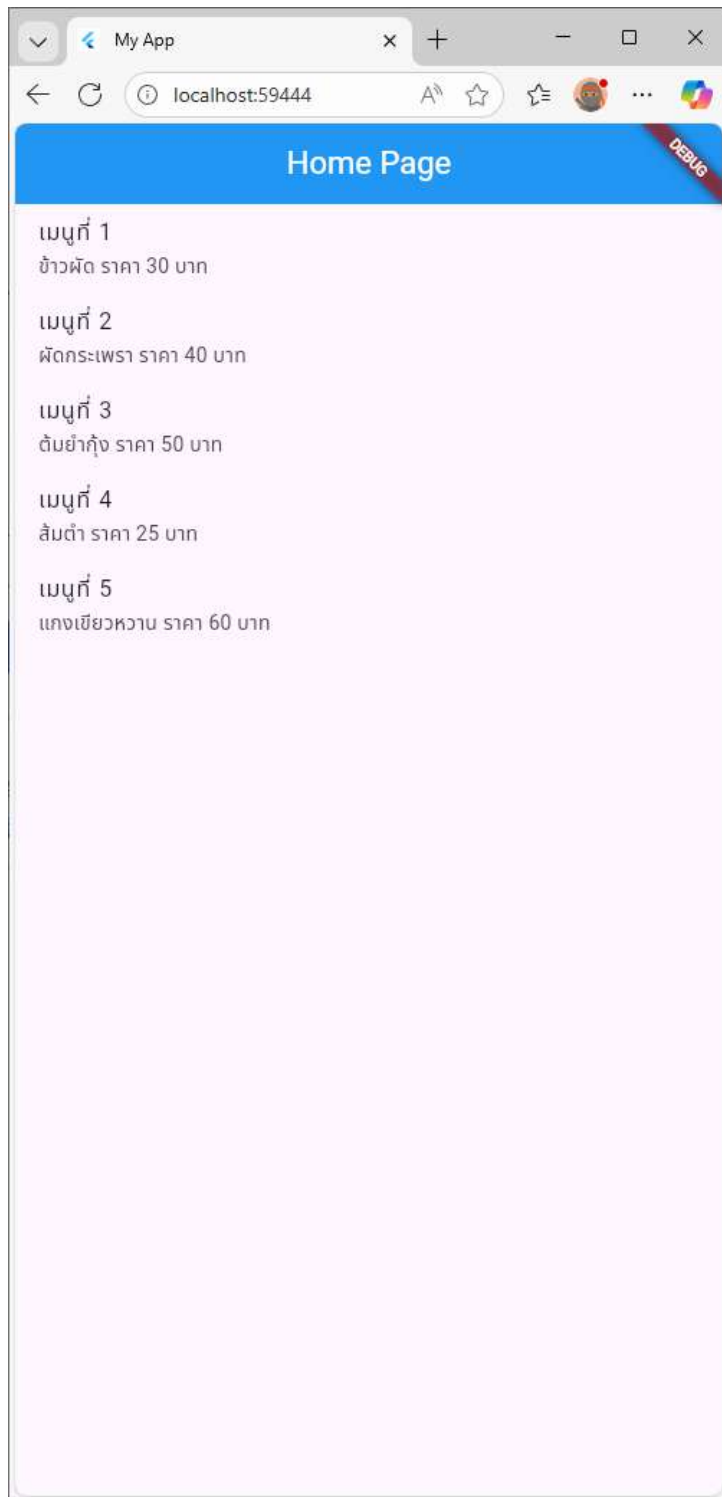
3

```

121   ),
122   SizedBox(height: 10),
123   Row(
124     mainAxisAlignment: .center,
125     children: [
126       calcBtn("4", () => pressNumber("4")),
127       calcBtn("5", () => pressNumber("5")),
128       calcBtn("6", () => pressNumber("6")),
129       calcBtn("-", () => pressOperator("-")),
130     ],
131   ),
132   SizedBox(height: 10),
133   Row(
134     mainAxisAlignment: .center,
135     children: [
136       calcBtn("1", () => pressNumber("1")),
137       calcBtn("2", () => pressNumber("2")),
138       calcBtn("3", () => pressNumber("3")),
139       calcBtn("*", () => pressOperator("*")),
140     ],
141   ),
142   SizedBox(height: 10),
143   Row(
144     mainAxisAlignment: .center,
145     children: [
146       calcBtn("0", () => pressNumber("0")),
147       calcBtn("C", () => clear()),
148       calcBtn("=", () => calculate()),
149       calcBtn("/", () => pressOperator("/")),
150     ],
151   ),
152   ],
153   ),
154   ),
155   );
156 }
157 }
158

```

4



5

CW3

6

```

1 // main.dart
2 import 'package:flutter/material.dart';
3 import 'foodMenu.dart';
4
5 void main() {
6   runApp(const MyApp());
7 }
8
9 class MyApp extends StatelessWidget {
10   const MyApp({super.key});
11
12   @override
13   Widget build(BuildContext context) {
14     return const MaterialApp(home: MyHomePage(), title: 'My App');
15   }
16 }
17
18 class MyHomePage extends StatefulWidget {
19   const MyHomePage({super.key});
20
21   @override
22   State<MyHomePage> createState() => _MyHomePageState();
23 }
24
25 class _MyHomePageState extends State<MyHomePage> {
26   // int count = 0;
27
28   // List<Widget> getData(int count) {
29   //   List<Widget> data = [];
30
31   //   data.add(Text('กดปุ่มเพื่อเพิ่มตัวเลข', style: TextStyle(fontSize: 32)));
32   //   data.add(Text(count.toString(), style: TextStyle(fontSize: 32)));
33
34   //   for (var i = 0; i < count; i++) {
35   //     var menu = ListTile(
36   //       title: Text(
37   //         "เมนูที่ ${i + 1}",
38   //         style: TextStyle(color: Colors.blue, fontSize: 20),
39   //       ),
40   //       subtitle: Text("รายละเอียดเมนูที่ ${i + 1}"),
41   //     );
42   //     data.add(menu);
43   //   }
44   //   return data;
45   // }
46
47   List<FoodMenu> menu = [
48     FoodMenu(foodname: "ข้าวผัด", foodprice: "30"),
49     FoodMenu(foodname: "ผัดกระเพรา", foodprice: "40"),
50     FoodMenu(foodname: "ต้มยำกุ้ง", foodprice: "50"),
51     FoodMenu(foodname: "ส้มตำ", foodprice: "25"),
52     FoodMenu(foodname: "แกงเขียวหวาน", foodprice: "60"),
53   ];
54
55   @override
56   Widget build(BuildContext context) {
57     return Scaffold(
58       appBar: AppBar(

```

```

59     title: Center(
60       child: Text('Home Page', style: TextStyle(color: Colors.white)),
61     ),
62     backgroundColor: Colors.blue,
63   ),
64   body: ListView.builder(
65     itemBuilder: (BuildContext context, int index) {
66       FoodMenu food = menu[index];
67       return ListTile(
68         title: Text("เมนูที่ ${index + 1}"),
69         subtitle: Text("${food.foodname} ราคา ${food.foodprice} บาท"),
70       );
71     },
72     itemCount: menu.length,
73   ),
74   // floatingActionButton: Column(
75   //   mainAxisAlignment: MainAxisAlignment.end,
76   //   children: [
77   //     FloatingActionButton(
78   //       onPressed: () {
79   //         setState(() {
80   //           if (count > 0) count -= 1;
81   //         });
82   //       },
83   //       child: Icon(Icons.remove),
84   //       heroTag: 'decrease',
85   //     ),
86   //     SizedBox(height: 10),
87   //     FloatingActionButton(
88   //       onPressed: () {
89   //         setState(() {
90   //           count += 1;
91   //         });
92   //       },
93   //       child: Icon(Icons.add),
94   //       heroTag: 'increase',
95   //     ),
96   //   ],
97   // ),
98 );
99 }
100 }
101
102 // foodMenu.dart
103 class FoodMenu {
104   String foodname;
105   String foodprice;
106
107   FoodMenu({required this.foodname, required this.foodprice});
108 }

```

7

CW4

8

```

1  import 'package:flutter/material.dart';
2  import 'food_menu.dart';
3
4  void main() {
5    runApp(const MyApp());
6  }
7
8  class MyApp extends StatelessWidget {
9    const MyApp({super.key});
10
11    @override
12    Widget build(BuildContext context) {
13      return const MaterialApp(home: MyHomePage(), title: 'My App');
14    }
15  }
16
17  class MyHomePage extends StatefulWidget {
18    const MyHomePage({super.key});
19
20    @override
21    State<MyHomePage> createState() => _MyHomePageState();
22  }
23
24  class _MyHomePageState extends State<MyHomePage> {
25    List<Map<String, dynamic>> listMenu = [];
26    // int count = 0;
27
28    // List<Widget> getData(int count) {
29    //   List<Widget> data = [];
30
31    //   data.add(Text('กดปุ่มเพื่อเพิ่มตัวเลข', style: TextStyle(fontSize: 32)));
32    //   data.add(Text(count.toString(), style: TextStyle(fontSize: 32)));
33
34    //   for (var i = 0; i < count; i++) {
35    //     var menu = ListTile(
36    //       title: Text(
37    //         "เมนูที่ ${i + 1}",
38    //         style: TextStyle(color: Colors.blue, fontSize: 20),
39    //       ),
40    //       subtitle: Text("รายละเอียดเมนูที่ ${i + 1}"),
41    //     );
42    //     data.add(menu);
43    //   }
44    //   return data;
45    // }
46
47    List<FoodMenu> menu = [
48      FoodMenu(
49        foodname: "กระเพราหมู",
50        foodprice: "50",
51        img: "assets/images/m1.jpg",
52        type: "ผัด",
53      ),
54      FoodMenu(
55        foodname: "ผัดไทยกุ้ง",
56        foodprice: "60",
57        img: "assets/images/m2.jpg",
58        type: "ผัด",

```

```

59     ),
60     FoodMenu(
61       foodname: "ซูชิ",
62       foodprice: "80",
63       img: "assets/images/m3.jpg",
64       type: "ญี่ปุ่น",
65     ),
66     FoodMenu(
67       foodname: "บาร์บีคิว",
68       foodprice: "150",
69       img: "assets/images/m4.jpg",
70       type: "ย่าง",
71     ),
72     FoodMenu(
73       foodname: "ชาลาเปา",
74       foodprice: "40",
75       img: "assets/images/m5.jpg",
76       type: "นึ่ง",
77     ),
78     FoodMenu(
79       foodname: "กึ่งเผา",
80       foodprice: "200",
81       img: "assets/images/m6.jpg",
82       type: "เผา",
83     ),
84     FoodMenu(
85       foodname: "ต้มมะระ",
86       foodprice: "70",
87       img: "assets/images/m7.jpg",
88       type: "ต้ม",
89     ),
90     FoodMenu(
91       foodname: "ปลาทอด",
92       foodprice: "120",
93       img: "assets/images/m8.jpg",
94       type: "ทอด",
95     ),
96     FoodMenu(
97       foodname: "แฮมมเนียง",
98       foodprice: "90",
99       img: "assets/images/m9.jpg",
100      type: "ย่าง",
101    ),
102    FoodMenu(
103      foodname: "เบอร์เกอร์",
104      foodprice: "110",
105      img: "assets/images/m10.jpg",
106      type: "นึ่ง",
107    ),
108    FoodMenu(
109      foodname: "สปาเก็ตตี้",
110      foodprice: "70",
111      img: "assets/images/m11.jpg",
112      type: "ต้ม",
113    ),
114    FoodMenu(
115      foodname: "พิซซ่า",
116      foodprice: "130",
117      img: "assets/images/m12.jpg",
118      type: "อบ",
119    ),
120    FoodMenu(

```

9

```

121      foodname: "สลัดผัก",
122      foodprice: "60",
123      img: "assets/images/m13.jpg",
124      type: "สด",
125    ),
126    FoodMenu(
127      foodname: "ข้าวมันไก่",
128      foodprice: "55",
129      img: "assets/images/m14.jpg",
130      type: "ต้ม",
131    ),
132    FoodMenu(
133      foodname: "ต้มยำกุ้ง",
134      foodprice: "80",
135      img: "assets/images/m15.jpg",
136      type: "ต้ม",
137    ),
138    FoodMenu(
139      foodname: "ข้าวผัด",
140      foodprice: "50",
141      img: "assets/images/m16.jpg",
142      type: "ผัด",
143    ),
144    FoodMenu(
145      foodname: "น้ำพองไก่",
146      foodprice: "30",
147      img: "assets/images/m17.jpg",
148      type: "ทอด",
149    ),
150    FoodMenu(
151      foodname: "น้ำผลไม้ปั่น",
152      foodprice: "45",
153      img: "assets/images/m18.jpg",
154      type: "เครื่องดื่ม",
155    ),
156    FoodMenu(
157      foodname: "ชาเย็น",
158      foodprice: "40",
159      img: "assets/images/m19.jpg",
160      type: "เครื่องดื่ม",
161    ),
162    FoodMenu(
163      foodname: "กาแฟเย็น",
164      foodprice: "50",
165      img: "assets/images/m20.jpg",
166      type: "เครื่องดื่ม",
167    ),
168  ];
169
170  @override
171  Widget build(BuildContext context) {
172    return Scaffold(
173      appBar: AppBar(
174        title: Center(
175          child: Text('Home Page', style: TextStyle(color: Colors.white)),
176        ),
177        backgroundColor: Colors.blue,
178      ),
179      body: ListView.builder(
180        itemBuilder: (BuildContext context, int index) {
181          FoodMenu food = menu[index];
182          return ListTile(

```

10

```

183 leading: Image.asset(food.img, width: 70, fit: BoxFit.cover),
184 title: Text("เมนูที่ ${index + 1} ${food.foodname}"),
185 subtitle: Text(
186   "ประเภทเมนู ${food.type} ราคา ${food.foodprice} บาท",
187 ),
188 onTap: () {
189   setState(() {
190     listMenu.add({
191       'food_name': food.foodname,
192       'food_price': food.foodprice,
193     });
194   });
195   showDialog(
196     context: context,
197     builder: (BuildContext context) {
198       return AlertDialog(
199         title: Text(
200           "คุณได้เพิ่ม ${food.foodname} ลงในรายการแล้ว",
201           style: TextStyle(fontSize: 20),
202         ),
203         content: SizedBox(
204           height: 500,
205           child: listMenu.isEmpty
206             ? const Center(child: Text('ยังไม่มีเมนูในตะกร้า'))
207             : SingleChildScrollView(
208               child: Column(
209                 mainAxisAlignment: MainAxisAlignment.min,
210                 children: [
211                   Text(
212                     "จำนวนเมนูทั้งหมด: ${listMenu.length} รายการ",
213                     style: TextStyle(
214                       fontSize: 16,
215                       fontWeight: FontWeight.bold,
216                     ),
217                   ),
218                   SizedBox(height: 10),
219                   ...listMenu.map((item) {
220                     return ListTile(
221                       dense: true,
222                       title: Text(item['food_name']!),
223                       trailing: Text(
224                         "${item['food_price']} บาท",
225                       ),
226                     );
227                   }),
228                   const Divider(),
229                   Row(
230                     mainAxisAlignment:
231                       MainAxisAlignment.spaceBetween,
232                     children: [
233                       const Text(
234                         'รวมทั้งสิ้น:',
235                         style: TextStyle(
236                           fontSize: 16,
237                           fontWeight: FontWeight.bold,
238                         ),
239                     ),
240                       Text(
241                         '${listMenu.fold<int>(0, (sum, item) => sum + int.parse(item['food
242 _price'].toString()))} บาท',
243                       style: const TextStyle(
244                         fontSize: 16,

```

11

```

244         fontWeight: FontWeight.bold,
245         color: Colors.blue,
246       ),
247     ),
248   ],
249 ),
250 ],
251 ),
252 ),
253 ),
254 // actions: [
255 //   TextButton(
256 //     child: const Text("ยกเลิก"),
257 //     onPressed: () => Navigator.pop(context),
258 //   ),
259 // ],
260 );
261 },
262 );
263 },
264 );
265 },
266 itemCount: menu.length,
267 ),
268 );
269 }
270 }
271

```

12



13

14

```

1 // main.dart
2 import 'package:flutter/material.dart';
3 import 'money_box.dart';
4
5 void main() {
6   runApp(const MyApp());
7 }
8
9 class MyApp extends StatelessWidget {
10   const MyApp({super.key});
11   @override
12   Widget build(BuildContext context) {
13     return MaterialApp(title: 'Container Demo', home: const MyHomePage());
14   }
15 }
16
17 class InputDecoratorExample extends StatelessWidget {
18   const InputDecoratorExample({super.key});
19
20   @override
21   Widget build(BuildContext context) {
22     return TextFormField(
23       decoration: InputDecoration(
24         border: const OutlineInputBorder(),
25         labelText: 'ชื่อเจ้าของบัญชี',
26         labelStyle: WidgetStateTextStyle.resolveWith((Set<WidgetState> state) {
27           final Color color = state.contains(WidgetState.error)
28             ? Theme.of(context).colorScheme.error
29             : Colors.blue;
30           return TextStyle(color: color, letterSpacing: 1.3);
31         })),
32     ),
33     validator: (String? value) {
34       if (value == null || value.isEmpty) {
35         return 'Enter account name';
36       }
37       return null;
38     },
39     autovalidateMode: AutovalidateMode.always,
40   );
41 }
42
43
44 class MyHomePage extends StatefulWidget {
45   const MyHomePage({super.key});
46   @override
47   State<MyHomePage> createState() => _MyHomePageState();
48 }
49
50 class _MyHomePageState extends State<MyHomePage> {
51   double outcome = 4000000.265614;
52   double income = 35000.54684;
53
54   @override
55   Widget build(BuildContext context) {
56     return Scaffold(
57       appBar: AppBar(
58         title: const Text('Container Example'),

```

```

59 // backgroundColor: Colors.blue,
60 ),
61 backgroundColor: Colors.amber,
62 body: SafeArea(
63 // ถ้ามี appBar ไม่ต้องใส่ SafeArea ก็ได้
64 child: Padding(
65 padding: const EdgeInsets.all(20),
66 child: Column(
67 spacing: 20,
68 children: [
69 Container(
70 margin: EdgeInsets.symmetric(vertical: 5, horizontal: 10),
71 height: 120,
72 padding: const EdgeInsets.all(20.0),
73 decoration: BoxDecoration(
74 color: Colors.white,
75 borderRadius: BorderRadius.circular(20),
76 ),
77 child: InputDecoratorExample(),
78 ),
79 MoneyBox(
80 title: "ยอดเงินคงเหลือ",
81 amount: (income - outcome) < 0 ? 0.0 : (income - outcome),
82 containerHeight: 100,
83 containerColor: Colors.green,
84 containerBorderRadius: 20,
85 ),
86 MoneyBox(
87 title: "รายได้",
88 amount: income,
89 containerHeight: 100,
90 containerColor: Colors.red,
91 containerBorderRadius: 20,
92 ),
93 MoneyBox(
94 title: "รายจ่าย",
95 amount: outcome,
96 containerHeight: 100,
97 containerColor: Colors.blue,
98 containerBorderRadius: 20,
99 ),
100 MoneyBox(
101 title: "ค่าจ่าย",
102 amount: (outcome - income) < 0 ? 0.0 : (outcome - income),
103 containerHeight: 100,
104 containerColor: Colors.brown,
105 containerBorderRadius: 20,
106 ),
107 TextButton(
108 onPressed: () {},
109 style: TextButton.styleFrom(
110 backgroundColor: Colors.green,
111 padding: EdgeInsets.symmetric(vertical: 10, horizontal: 20),
112 ),
113 child: const Text(
114 'Submit',
115 style: TextStyle(fontSize: 30, color: Colors.white),
116 ),
117 ),
118 ],
119 ),
120 ),

```

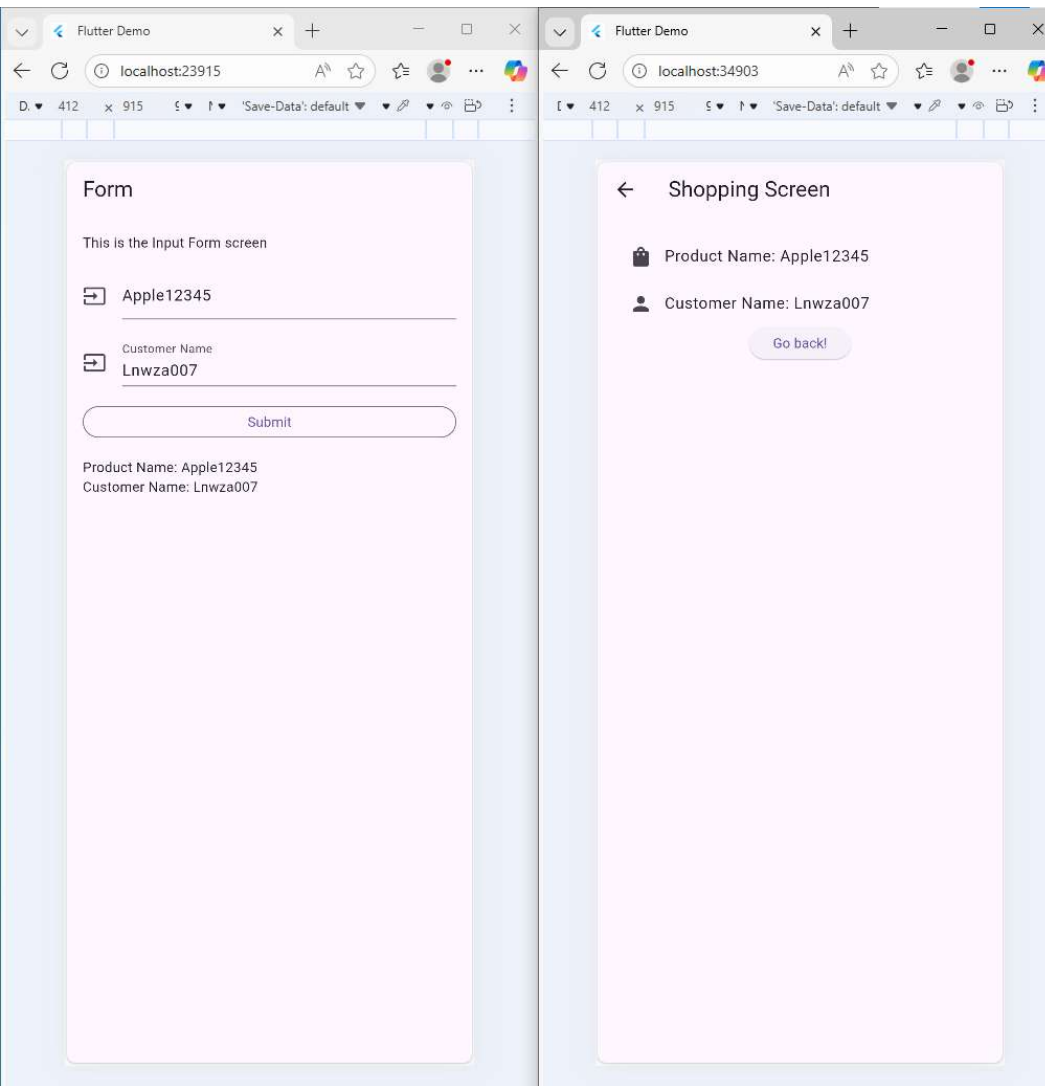
15

```

121 ),
122 );
123 }
124 }
125
126 // money_box.dart
127 import 'package:flutter/material.dart';
128 import 'package:intl/intl.dart';
129
130 class MoneyBox extends StatelessWidget {
131   const MoneyBox({
132     super.key,
133     required this.title,
134     required this.amount,
135     required this.containerHeight,
136     required this.containerColor,
137     required this.containerBorderRadius,
138   });
139   final TextStyle textStyleInContainer = const TextStyle(
140     color: Colors.white,
141     fontWeight: FontWeight.bold,
142     fontSize: 24,
143   );
144
145   final String title;
146   final double amount;
147   final double containerHeight;
148   final Color containerColor;
149   final double containerBorderRadius;
150
151   @override
152   Widget build(BuildContext context) {
153     return Container(
154       decoration: BoxDecoration(
155         color: containerColor,
156         borderRadius: BorderRadius.circular(containerBorderRadius),
157       ),
158       // width: 300,
159       height: containerHeight,
160       padding: EdgeInsets.all(20),
161       child: Row(
162         crossAxisAlignment: CrossAxisAlignment.end,
163         children: [
164           Text(title, style: textStyleInContainer),
165           Expanded(
166             child: Text(
167               NumberFormat("#,###.###").format(amount),
168               style: textStyleInContainer,
169               textAlign: TextAlign.end,
170             ),
171           ),
172         ],
173       ),
174     );
175   }
176 }

```

16



```

1 // main.dart
2 import 'package:flutter/material.dart';
3 import 'input_form.dart';
4
5 void main() {
6   runApp(const MyApp());
7 }
8
9 class MyApp extends StatefulWidget {
10   const MyApp({super.key});
11
12   @override
13   State<MyApp> createState() => _MainAppState();
14 }
15
16 class _MainAppState extends State<MyApp> {
17   @override
18   Widget build(BuildContext context) {
19     return const MaterialApp(
20       debugShowCheckedModeBanner: false,
21       title: 'Flutter Demo',
22       themeMode: ThemeMode.system,
23       home: InputForm(),
24     );
25   }
26 }
27
28 // input_form.dart
29 import 'package:flutter/material.dart';
30 import 'shopping_page.dart';
31
32 class InputForm extends StatefulWidget {
33   const InputForm({super.key});
34
35   @override
36   State<InputForm> createState() => _InputFormState();
37 }
38
39 class _InputFormState extends State<InputForm> {
40   String? _productName;
41   String? _customerName;
42
43   final _proController = TextEditingController();
44   final _cusController = TextEditingController();
45
46   void _updateText(String name) {
47     setState(() {
48       _productName = _proController.text;
49       _customerName = _cusController.text;
50     });
51   }
52
53   @override
54   void initState() {
55     super.initState();
56     _proController.addListener(() {
57       _updateText(_proController.text);
58     });

```

```

59     _cusController.addListener(() {
60       _updateText(_cusController.text);
61     });
62   }
63
64   @override
65   Widget build(BuildContext context) {
66     return Scaffold(
67       appBar: AppBar(title: Text('Form')),
68       body: Container(
69         padding: EdgeInsets.all(16.0),
70         child: ListView(
71           children: [
72             Text('This is the Input Form screen'),
73             SizedBox(height: 20),
74             TextField(
75               controller: _proController,
76               // onChanged: (value) => _updateText(value),
77               decoration: InputDecoration(
78                 hintText: 'Product Name',
79                 icon: Icon(Icons.input),
80               ),
81             ),
82             SizedBox(height: 20),
83             TextFormField(
84               controller: _cusController,
85               // onChanged: (value) => _updateText(value),
86               decoration: InputDecoration(
87                 labelText: 'Customer Name',
88                 icon: Icon(Icons.input),
89               ),
90             ),
91             SizedBox(height: 20),
92             OutlinedButton(
93               onPressed: () {
94                 Navigator.push(
95                   context,
96                   MaterialPageRoute(
97                     builder: (context) => FormShopping(
98                       productName: _productName ?? '',
99                       customerName: _customerName ?? '',
100                     ),
101                   ),
102               );
103             },
104             child: const Text('Submit'),
105           ),
106           SizedBox(height: 20),
107           Text("Product Name: ${_proController.text}"),
108           Text("Customer Name: ${_cusController.text}"),
109         ],
110       ),
111     );
112   }
113 }
114
115 // shopping_page.dart
116 import 'package:flutter/material.dart';
117
118 class FormShopping extends StatelessWidget {
119   const FormShopping({super.key, required this.productName, required this.customerName});

```

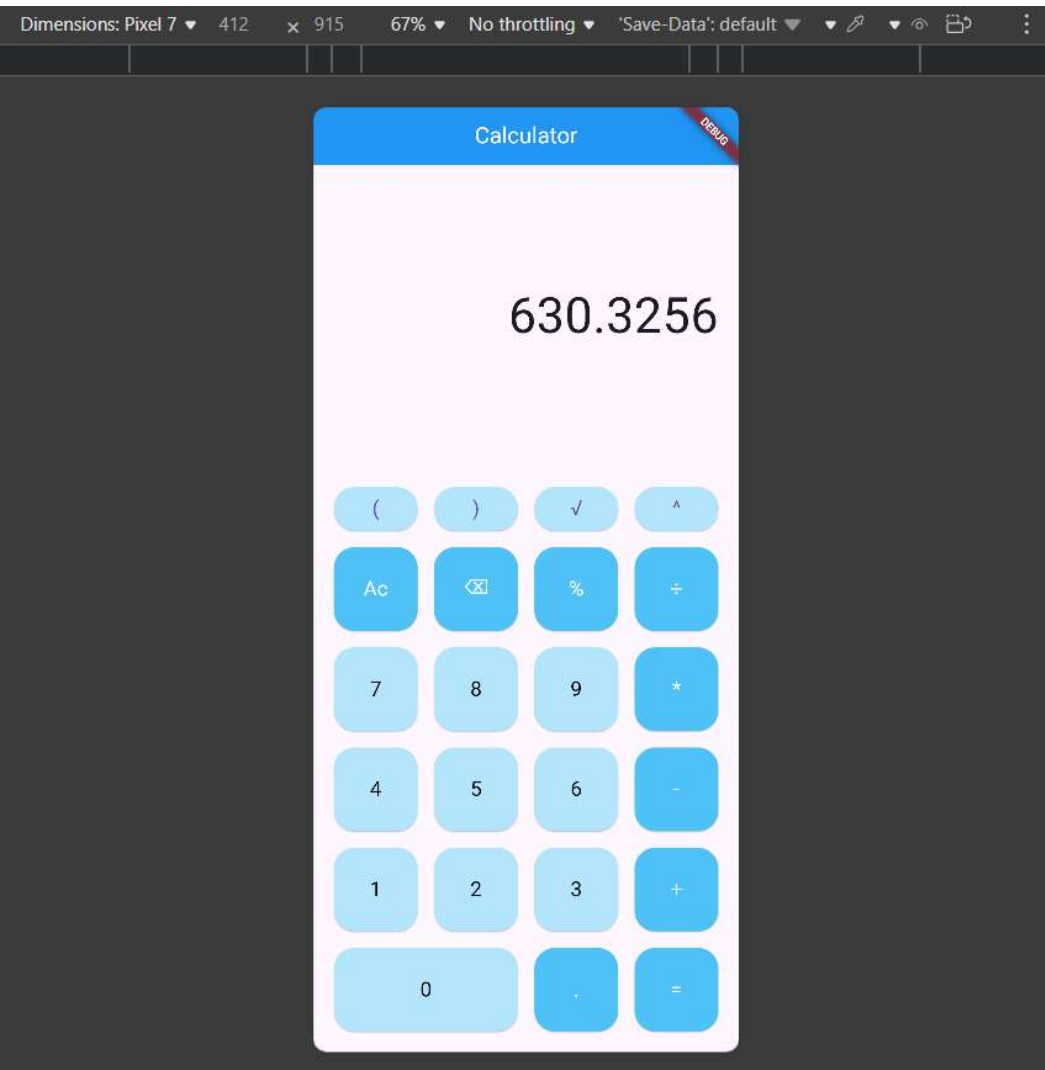
19

```

121
122   final String productName;
123   final String customerName;
124
125   @override
126   Widget build(BuildContext context) {
127     return Scaffold(
128       appBar: AppBar(title: Text('Shopping Screen')),
129       body: Container(
130         padding: EdgeInsets.all(16.0),
131         child: ListView(
132           children: [
133             ListTile(
134               leading: Icon(Icons.shopping_bag),
135               title: Text('Product Name: $productName'),
136             ),
137             ListTile(
138               leading: Icon(Icons.person),
139               title: Text('Customer Name: $customerName'),
140             ),
141             Center(
142               child: ElevatedButton(
143                 onPressed: () {
144                   Navigator.pop(context);
145                 },
146                 child: Text('Go back!'),
147               ),
148             ),
149           ],
150         ),
151       ),
152     );
153   }
154 }

```

20



```

1 import 'package:flutter/material.dart';
2 import 'package:flutter_staggered_grid_view/flutter_staggered_grid_view.dart';
3 import 'package:math_expressions/math_expressions.dart';
4
5 void main() {
6   runApp(const MyApp());
7 }
8 class MyApp extends StatelessWidget {
9   const MyApp({super.key});
10
11   @override
12   Widget build(BuildContext context) {
13     return MaterialApp(title: 'Calculator', home: MyHomePage());
14   }
15 }
16 class MyHomePage extends StatefulWidget {
17   const MyHomePage({super.key});
18
19   @override
20   State<MyHomePage> createState() => _MyHomePageState();
21 }
22 class _MyHomePageState extends State<MyHomePage> {
23   String result = "0";
24   final Set<String> operators = {'+', '-', '*', '/', '^'};
25
26   @override
27   Widget build(BuildContext context) {
28     return Scaffold(
29       appBar: AppBar(
30         title: Center(
31           child: const Text(
32             'Calculator',
33             style: TextStyle(color: Colors.white, fontWeight: FontWeight.w500),
34           ),
35         ),
36       backgroundColor: Colors.blue,
37     ),
38     body: Column(
39       children: [
40         Expanded(child: DisplaySection(displayText: result)),
41         Padding(
42           padding: const EdgeInsets.all(20.0),
43           child: BtnSection(
44             onPressed: (symbol) {
45               setState(() {
46                 if (result == 'Error') {
47                   // reset on any input after error
48                   result = '0';
49                 }
50                 if (symbol == 'Ac') {
51                   result = '0';
52                   return;
53                 }
54
55                 if (symbol == 'C') {
56                   if (result.length <= 1) {
57                     result = '0';
58                     return;

```

```

59     }
60     final lastChar = result[result.length - 1];
61     if (lastChar == '.') {
62         int lastOp = -1;
63         for (int i = result.length - 2; i >= 0; i--) {
64             if (operators.contains(result[i])) {
65                 lastOp = i;
66                 break;
67             }
68         }
69         final currentNumber = result.substring(
70             lastOp + 1,
71             result.length - 1,
72         );
73         if (currentNumber == '0') {
74             // remove the '0.' together
75             result = result.substring(0, result.length - 2);
76             if (result.isEmpty) result = '0';
77         } else {
78             // remove only the dot
79             result = result.substring(0, result.length - 1);
80             if (result.isEmpty) result = '0';
81         }
82         return;
83     }
84     // If deleting a 'V' that was inserted after a '(' (e.g. "√(√)",
85     // remove the '(' together with the trailing '√' so it becomes "√".
86     if (lastChar == '√') {
87         if (result.length >= 2 &&
88             result[result.length - 2] == '(') {
89             result = result.substring(0, result.length - 2);
90             if (result.isEmpty) result = '0';
91             return;
92         }
93     }
94     // normal backspace
95     result = result.substring(0, result.length - 1);
96     if (result.isEmpty) result = '0';
97     return;
98 }
99
100 if (symbol == '=') {
101     try {
102         String expr = result;
103         expr = expr.replaceAll('/', '/');
104         expr = expr.replaceAll('x', '*');
105
106         expr = expr.replaceAllMapped(
107             RegExp(r"(\d+(?:\.\d+)?)%"),
108             (Match m) => '(${m.group(1)}/100)',
109         );
110         expr = expr.replaceAll('√', 'sqrt(');
111         expr = expr.replaceAllMapped(
112             RegExp(r'√(\d+(?:\.\d+)?)'),
113             (Match m) => 'sqrt(${m.group(1)})',
114         );
115
116         while (expr.isNotEmpty &&
117             RegExp(r'[+|-/*^]').hasMatch(expr[expr.length - 1])) {
118             expr = expr.substring(0, expr.length - 1);
119         }
120     }

```

23

```

121     GrammarParser p = GrammarParser();
122     Expression exp = p.parse(expr);
123     ContextModel cm = ContextModel();
124     var evaluator = RealEvaluator(cm);
125     final double value = evaluator.evaluate(exp).toDouble();
126     if (value.isInfinite || value.isNaN) {
127         result = 'Error';
128     } else if (value % 1 == 0) {
129         result = value.toInt().toString();
130     } else {
131         // Round to 4 decimal places, then trim trailing zeros.
132         double rounded =
133             (value * 10000).roundToDouble() / 10000.0;
134         String s = rounded.toStringAsFixed(
135             4,
136         ); // ensures 4 decimal places
137         // Remove trailing zeros and optional trailing dot
138         s = s.replaceFirst(RegExp(r"\.?.?0+\$"), '');
139         result = s;
140     }
141 } catch (e) {
142     result = 'Error';
143 }
144 return;
145 }
146
147 if (symbol == '.') {
148     int lastOp = -1;
149     for (int i = result.length - 1; i >= 0; i--) {
150         if (operators.contains(result[i])) {
151             lastOp = i;
152             break;
153         }
154     }
155     final currentNumber = result.substring(lastOp + 1);
156     if (currentNumber.contains('.')) {
157         // already has a decimal point in current number – do nothing
158         return;
159     } else if (currentNumber.isEmpty) {
160         // last char was an operator (or result was empty) – add 0.
161         result = '${result}0.';
162         return;
163     } else if (result == '0') {
164         // replace initial zero with a decimal number
165         result = '0.';
166         return;
167     } else {
168         result = '$result.';
169         return;
170     }
171 }
172
173 if (symbol == '√') {
174     int lastOp = -1;
175     for (int i = result.length - 1; i >= 0; i--) {
176         if (operators.contains(result[i])) {
177             lastOp = i;
178             break;
179         }
180     }
181     final currentNumber = result.substring(lastOp + 1);
182     if (currentNumber == '0') {

```

24

```

183         result = '${result.substring(0, lastOp + 1)}v';
184         if (result.isEmpty) result = 'v';
185         return;
186     } else if (result == '0') {
187         result = 'v';
188         return;
189     } else {
190         // If the last character is already a 'v', insert a '(' before
191         // appending the new 'v' so sequences like `vv` become `v(v`.
192         final lastChar = result[result.length - 1];
193         if (lastChar == 'v') {
194             result = '$result(v';
195         } else {
196             result = '$resultv';
197         }
198         return;
199     }
200 }
201
202 if (operators.contains(symbol)) {
203     final lastChar = result[result.length - 1];
204     if (operators.contains(lastChar)) {
205         result = result.substring(0, result.length - 1) + symbol;
206     } else {
207         result = result + symbol;
208     }
209     return;
210 }
211
212 if (symbol == '%') {
213     // Disallow '%' immediately after an operator, a dot, a 'v', or another '%'.
214     final lastChar = result[result.length - 1];
215     if (operators.contains(lastChar) ||
216         lastChar == '.' ||
217         lastChar == 'v') {
218         // ignore invalid placement
219         return;
220     }
221 }
222
223 // Also prevent putting multiple '%' in the same number segment
224 int lastOp = -1;
225 for (int i = result.length - 1; i >= 0; i--) {
226     if (operators.contains(result[i])) {
227         lastOp = i;
228         break;
229     }
230 }
231 final currentNumber = result.substring(lastOp + 1);
232 if (currentNumber.contains('%')) return;
233
234 result = '$result%';
235 return;
236 }
237
238 // Default: append digits/symbols
239 if (result == '0') {
240     result = symbol;
241 } else {
242     result = result + symbol;
243 }
244 },

```

25

```

245     ),
246     ),
247     ],
248     ),
249     );
250 }
251 }
252 class DisplaySection extends StatefulWidget {
253     final String displayText;
254     const DisplaySection({super.key, required this.displayText});
255
256     @override
257     State<DisplaySection> createState() => _DisplaySectionState();
258 }
259 class _DisplaySectionState extends State<DisplaySection> {
260     final ScrollController _scrollController = ScrollController();
261
262     @override
263     void didUpdateWidget(covariant DisplaySection oldWidget) {
264         super.didUpdateWidget(oldWidget);
265         if (oldWidget.displayText != widget.displayText) {
266             // Wait for the frame so the new text layout is applied, then scroll to end.
267             WidgetsBinding.instance.addPostFrameCallback((_) {
268                 if (_scrollController.hasClients) {
269                     _scrollController.animateTo(
270                         _scrollController.position.maxScrollExtent,
271                         duration: const Duration(milliseconds: 200),
272                         curve: Curves.easeOut,
273                     );
274                 }
275             });
276         }
277     }
278
279     @override
280     void dispose() {
281         _scrollController.dispose();
282         super.dispose();
283     }
284
285     @override
286     Widget build(BuildContext context) {
287         // Horizontal scroll that stays right-aligned when content is short,
288         // and allows scrolling when the text is wider than the screen.
289         return Scrollbar(
290             controller: _scrollController,
291             thumbVisibility: true,
292             child: SingleChildScrollView(
293                 scrollDirection: Axis.horizontal,
294                 controller: _scrollController,
295                 child: ConstrainedBox(
296                     constraints: BoxConstraints(
297                         minWidth: MediaQuery.of(context).size.width,
298                     ),
299                 child: Container(
300                     padding: const EdgeInsets.all(20.0),
301                     alignment: Alignment.centerRight,
302                     child: Text(
303                         widget.displayText,
304                         style: const TextStyle(fontSize: 48, fontWeight: FontWeight.w500),
305                     ),
306                 ),

```

26

```

307     ),
308     ),
309     );
310 }
311 }
312
313 class BtnSection extends StatefulWidget {
314   final ValueChanged<String> onPressed;
315   const BtnSection({super.key, required this.onPressed});
316
317   @override
318   State<BtnSection> createState() => _BtnSectionState();
319 }
320 class _BtnSectionState extends State<BtnSection> {
321   static double spacingSize = 16.0;
322   static double borderRadius = 20.0;
323
324   @override
325   Widget build(BuildContext context) {
326     return StaggeredGrid.count(
327       crossAxisCount: 4,
328       mainAxisSpacing: spacingSize,
329       crossAxisSpacing: spacingSize,
330       children: [
331         StaggeredGridTile.count(
332           crossAxisCellCount: 1,
333           mainAxisCellCount: 0.6,
334           child: SmallBtn(symbol: "(", onPressed: widget.onPressed),
335         ),
336         StaggeredGridTile.count(
337           crossAxisCellCount: 1,
338           mainAxisCellCount: 0.6,
339           child: SmallBtn(symbol: ")", onPressed: widget.onPressed),
340         ),
341         StaggeredGridTile.count(
342           crossAxisCellCount: 1,
343           mainAxisCellCount: 0.6,
344           child: SmallBtn(symbol: "v", onPressed: widget.onPressed),
345         ),
346         StaggeredGridTile.count(
347           crossAxisCellCount: 1,
348           mainAxisCellCount: 0.6,
349           child: SmallBtn(symbol: "^", onPressed: widget.onPressed),
350         ),
351         StaggeredGridTile.count(
352           crossAxisCellCount: 1,
353           mainAxisCellCount: 1,
354           child: NormalBtn(
355             symbol: "Ac",
356             color: Colors.lightBlue[300]!,
357             textColor: Colors.white,
358             onPressed: widget.onPressed,
359           ),
360         ),
361         StaggeredGridTile.count(
362           crossAxisCellCount: 1,
363           mainAxisCellCount: 1,
364           child: NormalBtn(
365             symbol: "☒",
366             color: Colors.lightBlue[300]!,
367             textColor: Colors.white,
368             onPressed: widget.onPressed,

```

27

```

369     ),
370     ),
371     StaggeredGridTile.count(
372       crossAxisCellCount: 1,
373       mainAxisCellCount: 1,
374       child: NormalBtn(
375         symbol: "%",
376         color: Colors.lightBlue[300]!,
377         textColor: Colors.white,
378         onPressed: widget.onPressed,
379       ),
380     ),
381     StaggeredGridTile.count(
382       crossAxisCellCount: 1,
383       mainAxisCellCount: 1,
384       child: NormalBtn(
385         symbol: "÷",
386         color: Colors.lightBlue[300]!,
387         textColor: Colors.white,
388         onPressed: widget.onPressed,
389       ),
390     ),
391     StaggeredGridTile.count(
392       crossAxisCellCount: 1,
393       mainAxisCellCount: 1,
394       child: NormalBtn(
395         symbol: "7",
396         color: Colors.lightBlue[100]!,
397         textColor: Colors.black,
398         onPressed: widget.onPressed,
399       ),
400     ),
401     StaggeredGridTile.count(
402       crossAxisCellCount: 1,
403       mainAxisCellCount: 1,
404       child: NormalBtn(
405         symbol: "8",
406         color: Colors.lightBlue[100]!,
407         textColor: Colors.black,
408         onPressed: widget.onPressed,
409       ),
410     ),
411     StaggeredGridTile.count(
412       crossAxisCellCount: 1,
413       mainAxisCellCount: 1,
414       child: NormalBtn(
415         symbol: "9",
416         color: Colors.lightBlue[100]!,
417         textColor: Colors.black,
418         onPressed: widget.onPressed,
419       ),
420     ),
421     StaggeredGridTile.count(
422       crossAxisCellCount: 1,
423       mainAxisCellCount: 1,
424       child: NormalBtn(
425         symbol: "*",
426         color: Colors.lightBlue[300]!,
427         textColor: Colors.white,
428         onPressed: widget.onPressed,
429       ),
430     ),

```

28

```

431 StaggeredGridTile.count(
432   crossAxisCellCount: 1,
433   mainAxisCellCount: 1,
434   child: NormalBtn(
435     symbol: "4",
436     color: Colors.lightBlue[100]!,
437     textColor: Colors.black,
438     onPressed: widget.onPressed,
439   ),
440 ),
441 StaggeredGridTile.count(
442   crossAxisCellCount: 1,
443   mainAxisCellCount: 1,
444   child: NormalBtn(
445     symbol: "5",
446     color: Colors.lightBlue[100]!,
447     textColor: Colors.black,
448     onPressed: widget.onPressed,
449   ),
450 ),
451 StaggeredGridTile.count(
452   crossAxisCellCount: 1,
453   mainAxisCellCount: 1,
454   child: NormalBtn(
455     symbol: "6",
456     color: Colors.lightBlue[100]!,
457     textColor: Colors.black,
458     onPressed: widget.onPressed,
459   ),
460 ),
461 StaggeredGridTile.count(
462   crossAxisCellCount: 1,
463   mainAxisCellCount: 1,
464   child: NormalBtn(
465     symbol: "-",
466     color: Colors.lightBlue[300]!,
467     textColor: Colors.white,
468     onPressed: widget.onPressed,
469   ),
470 ),
471 StaggeredGridTile.count(
472   crossAxisCellCount: 1,
473   mainAxisCellCount: 1,
474   child: NormalBtn(
475     symbol: "1",
476     color: Colors.lightBlue[100]!,
477     textColor: Colors.black,
478     onPressed: widget.onPressed,
479   ),
480 ),
481 StaggeredGridTile.count(
482   crossAxisCellCount: 1,
483   mainAxisCellCount: 1,
484   child: NormalBtn(
485     symbol: "2",
486     color: Colors.lightBlue[100]!,
487     textColor: Colors.black,
488     onPressed: widget.onPressed,
489   ),
490 ),
491 StaggeredGridTile.count(
492   crossAxisCellCount: 1,

```

```

493   mainAxisCellCount: 1,
494   child: NormalBtn(
495     symbol: "3",
496     color: Colors.lightBlue[100]!,
497     textColor: Colors.black,
498     onPressed: widget.onPressed,
499   ),
500 ),
501 StaggeredGridTile.count(
502   crossAxisCellCount: 1,
503   mainAxisCellCount: 1,
504   child: NormalBtn(
505     symbol: "+",
506     color: Colors.lightBlue[300]!,
507     textColor: Colors.white,
508     onPressed: widget.onPressed,
509   ),
510 ),
511 StaggeredGridTile.count(
512   crossAxisCellCount: 2,
513   mainAxisCellCount: 1,
514   child: NormalBtn(
515     symbol: "0",
516     color: Colors.lightBlue[100]!,
517     textColor: Colors.black,
518     onPressed: widget.onPressed,
519   ),
520 ),
521 StaggeredGridTile.count(
522   crossAxisCellCount: 1,
523   mainAxisCellCount: 1,
524   child: NormalBtn(
525     symbol: ".",
526     color: Colors.lightBlue[300]!,
527     textColor: Colors.white,
528     onPressed: widget.onPressed,
529   ),
530 ),
531 StaggeredGridTile.count(
532   crossAxisCellCount: 1,
533   mainAxisCellCount: 1,
534   child: NormalBtn(
535     symbol: "=",
536     color: Colors.lightBlue[300]!,
537     textColor: Colors.white,
538     onPressed: widget.onPressed,
539   ),
540 ),
541   ],
542 );
543 }
544 }
545 class SmallBtn extends StatefulWidget {
546   final String symbol;
547   final ValueChanged<String>? onPressed;
548   const SmallBtn({super.key, required this.symbol, this.onPressed});
549
550   @override
551   State<SmallBtn> createState() => _SmallBtnState();
552 }
553
554 class _SmallBtnState extends State<SmallBtn> {

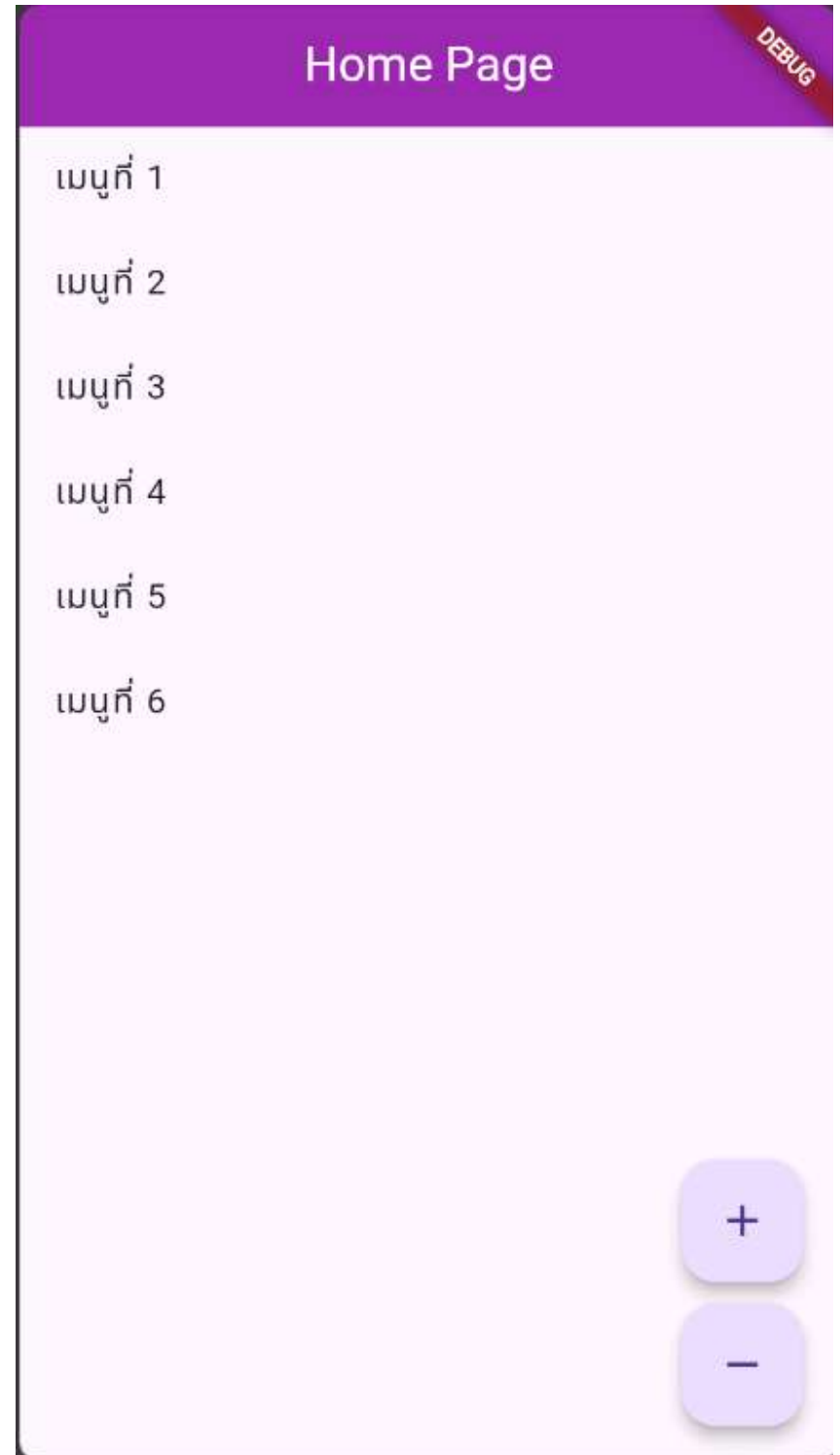
```

```

555 @override
556 Widget build(BuildContext context) {
557   return ElevatedButton(
558     style: ElevatedButton.styleFrom(
559       shape: RoundedRectangleBorder(
560         borderRadius: BorderRadius.circular(_BtnSectionState.borderRadius),
561       ),
562       backgroundColor: Colors.lightBlue[100],
563     ),
564     onPressed: () => widget.onPressed?.call(widget.symbol),
565     child: Text(widget.symbol, style: const TextStyle(fontSize: 20)),
566   );
567 }
568 }
569 class NormalBtn extends StatefulWidget {
570   final String symbol;
571   final Color color;
572   final Color textColor;
573   final ValueChanged<String> onPressed;
574   const NormalBtn({
575     super.key,
576     required this.symbol,
577     required this.color,
578     required this.textColor,
579     required this.onPressed,
580   });
581
582   @override
583   State<NormalBtn> createState() => _NormalBtnState();
584 }
585 class _NormalBtnState extends State<NormalBtn> {
586   @override
587   Widget build(BuildContext context) {
588     return ElevatedButton(
589       style: ElevatedButton.styleFrom(
590         shape: RoundedRectangleBorder(
591           borderRadius: BorderRadius.circular(_BtnSectionState.borderRadius),
592         ),
593         backgroundColor: widget.color,
594       ),
595       onPressed: () {
596         widget.onPressed(widget.symbol);
597       },
598       child: Text(
599         widget.symbol,
600         style: TextStyle(
601           color: widget.textColor,
602           fontSize: 20,
603           fontWeight: FontWeight.w500,
604         ),
605       ),
606     );
607   }
608 }
609

```

31



32

```

1 import 'package:flutter/material.dart';
2 void main() {
3   runApp(const MyApp());
4 }
5 class MyApp extends StatelessWidget {
6   const MyApp({super.key});
7   @override
8   Widget build(BuildContext context) {
9     return const MaterialApp(home: MyHomePage(), title: 'My App');
10  }
11 }
12 class MyHomePage extends StatefulWidget {
13   const MyHomePage({super.key});
14   @override
15   State<MyHomePage> createState() => _MyHomePageState();
16 }
17 class _MyHomePageState extends State<MyHomePage> {
18   int count = 0;
19   @override
20   Widget build(BuildContext context) {
21     return Scaffold(
22       appBar: AppBar(
23         title: Center(
24           child: Text('Home Page', style: TextStyle(color: Colors.white)),
25         ),
26       backgroundColor: Colors.purple,
27     ),
28     body: ListView.builder(
29       itemBuilder: (BuildContext context, int index) {
30         return ListTile(title: Text("เมนูที่ ${index + 1}"));
31       },
32       itemCount: count,
33     ),
34     floatingActionButton: Column(
35       mainAxisAlignment: .end,
36       children: [
37         FloatingActionButton(
38           onPressed: () {
39             setState(() { count += 1; });
40           },
41           heroTag: 'increase',
42           child: Icon(Icons.add),
43         ),
44         SizedBox(height: 10),
45         FloatingActionButton(
46           onPressed: () {
47             setState(() { if (count > 0) count -= 1; });
48           },
49           heroTag: 'decrease',
50           child: Icon(Icons.remove),
51         ),
52       ],
53     ),
54   );
55 }
56 }

```



## HW5 Application เลขท้าย 3 ตัว

```

1 // main.dart
2 import 'package:flutter/material.dart';
3 import 'homepage.dart';
4
5 void main() {
6   runApp(const MyApp());
7 }
8
9 class MyApp extends StatelessWidget {
10   const MyApp({super.key});
11
12   @override
13   Widget build(BuildContext context) {
14     return const MaterialApp(home: HomePage());
15   }
16 }
17
18 // homepage.dart
19 import 'package:flutter/material.dart';
20 import 'package:flutter/services.dart';
21 import 'dart:math';
22 import 'result.dart';
23
24 class HomePage extends StatefulWidget {
25   const HomePage({super.key});
26
27   @override
28   State<HomePage> createState() => _HomePageState();
29 }
30
31 class _HomePageState extends State<HomePage> {
32   final _firstDigitController = TextEditingController();
33   final _secondDigitController = TextEditingController();
34   final _thirdDigitController = TextEditingController();
35   final _amountController = TextEditingController();
36   final random = Random();
37
38   @override
39   Widget build(BuildContext context) {
40     return Scaffold(
41       body: SafeArea(
42         child: Padding(
43           padding: const EdgeInsets.all(50.0),
44           child: Column(
45             children: [
46               Row(
47                 spacing: 40.0,
48                 children: [
49                   Expanded(
50                     child: Column(
51                       spacing: 10.0,
52                       mainAxisAlignment: MainAxisAlignment.center,
53                       children: [
54                         Text('หลักที่ 3', style: TextStyle(fontSize: 18)),
55                         TextField(
56                           controller: _thirdDigitController,
57                           decoration: InputDecoration(
58                             border: OutlineInputBorder(),

```

35

```

59       ),
60       keyboardType: TextInputType.number,
61       inputFormatters: [
62         FilteringTextInputFormatter.digitsOnly,
63         LengthLimitingTextInputFormatter(1),
64       ],
65     ),
66   ],
67 ),
68 ),
69 Expanded(
70   child: Column(
71     spacing: 10.0,
72     mainAxisAlignment: MainAxisAlignment.center,
73     children: [
74       Text('หลักที่ 2', style: TextStyle(fontSize: 18)),
75       TextField(
76         controller: _secondDigitController,
77         decoration: InputDecoration(
78           border: OutlineInputBorder(),
79         ),
80         keyboardType: TextInputType.number,
81         inputFormatters: [
82           FilteringTextInputFormatter.digitsOnly,
83           LengthLimitingTextInputFormatter(1),
84         ],
85       ),
86     ],
87 ),
88 ),
89 Expanded(
90   child: Column(
91     spacing: 10.0,
92     mainAxisAlignment: MainAxisAlignment.center,
93     children: [
94       Text('หลักที่ 1', style: TextStyle(fontSize: 18)),
95       TextField(
96         controller: _firstDigitController,
97         decoration: InputDecoration(
98           border: OutlineInputBorder(),
99         ),
100        keyboardType: TextInputType.number,
101        inputFormatters: [
102          FilteringTextInputFormatter.digitsOnly,
103          LengthLimitingTextInputFormatter(1),
104        ],
105      ),
106    ],
107  ),
108 ),
109 ],
110 ),
111 SizedBox(height: 60.0),
112 Text("จำนวนเงินที่ต้องการซื้อ", style: TextStyle(fontSize: 24)),
113 SizedBox(height: 10.0),
114 Row(
115   children: [
116     Expanded(
117       child: TextField(
118         controller: _amountController,
119         decoration: InputDecoration(border: OutlineInputBorder()),
120         keyboardType: TextInputType.number,

```

36

```

121         inputFormatters: [FilteringTextInputFormatter.digitsOnly],
122     ),
123 ),
124 SizedBox(width: 20.0),
125 Text("บาท", style: TextStyle(fontSize: 24)),
126 ],
127 ),
128 Expanded(child: Container()),
129 Row(
130   children: [
131     Expanded(
132       child: OutlinedButton(
133         onPressed: () {
134           if (_firstDigitController.text.isEmpty ||
135             _secondDigitController.text.isEmpty ||
136             _thirdDigitController.text.isEmpty) {
137             ScaffoldMessenger.of(context).showSnackBar(
138               const SnackBar(
139                 content: Text('กรุณากรอกเลขทั้ง 3 หลัก'),
140               ),
141             );
142             return;
143           }
144           if (_amountController.text.isEmpty) {
145             ScaffoldMessenger.of(context).showSnackBar(
146               const SnackBar(content: Text('กรุณากรอกจำนวนเงิน')),
147             );
148             return;
149           }
150
151           Navigator.push(
152             context,
153             MaterialPageRoute(
154               builder: (context) => ResultPage(
155                 firstDigit: _firstDigitController.text,
156                 secondDigit: _secondDigitController.text,
157                 thirdDigit: _thirdDigitController.text,
158                 amount: int.parse(_amountController.text),
159                 random:
160                   "${random.nextInt(10)}${random.nextInt(10)}${random.nextInt(10)}",
161                 // random: '999',
162               ),
163             ),
164           );
165         },
166         style: OutlinedButton.styleFrom(
167           shape: RoundedRectangleBorder(
168             borderRadius: BorderRadius.circular(10.0),
169           ),
170           padding: EdgeInsets.symmetric(vertical: 20.0),
171         ),
172         child: Text(
173           "คลิก ตรวจสอบผลลัพธ์",
174           style: TextStyle(fontSize: 24),
175         ),
176       ),
177     ),
178   ],
179 ),
180 ],
181 ),
182 ),

```

37

```

183     ),
184   );
185 }
186 }
187
188 // result.dart
189 import 'package:flutter/material.dart';
190 import 'package:intl/intl.dart';
191
192 class ResultPage extends StatelessWidget {
193   const ResultPage({
194     super.key,
195     required this.firstDigit,
196     required this.secondDigit,
197     required this.thirdDigit,
198     required this.amount,
199     required this.random,
200   });
201
202   final String firstDigit;
203   final String secondDigit;
204   final String thirdDigit;
205   final int amount;
206   final String random;
207
208   String get purchasedNumber => thirdDigit + secondDigit + firstDigit;
209
210   @override
211   Widget build(BuildContext context) {
212     return Scaffold(
213       body: SafeArea(
214         child: Padding(
215           padding: const EdgeInsets.all(50.0),
216           child: Column(
217             spacing: 20.0,
218             children: [
219               Row(
220                 children: [
221                   Text(
222                     "เลขที่คุณซื้อคือ $purchasedNumber",
223                     style: TextStyle(fontSize: 18),
224                   ),
225                 ],
226               ),
227               Row(
228                 children: [
229                   Text(
230                     "จำนวนเงินที่คุณซื้อ คือ $amount บาท",
231                     style: TextStyle(fontSize: 18),
232                   ),
233                 ],
234               ),
235               Row(
236                 children: [
237                   Text(
238                     "** เลขที่ออก คือ $random",
239                     style: TextStyle(fontSize: 18),
240                   ),
241                 ],
242               ),
243               if (purchasedNumber == random)
244                 Column(

```

38

245  
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40

```

1 // main.dart
2 import 'package:flutter/material.dart';
3 import 'form.dart';
4
5 void main() {
6   runApp(const MyApp());
7 }
8
9 class MyApp extends StatelessWidget {
10   const MyApp({super.key});
11
12   @override
13   Widget build(BuildContext context) {
14     return const MaterialApp(home: FormPage());
15   }
16 }
17
18 // form.dart
19 import 'package:flutter/material.dart';
20 import 'package:flutter/services.dart';
21 import 'field_study_radio_btn.dart';
22 import 'result.dart';
23
24 class FormPage extends StatefulWidget {
25   const FormPage({super.key});
26
27   @override
28   State<FormPage> createState() => _FormPageState();
29 }
30
31 class _FormPageState extends State<FormPage> {
32   final _formKey = GlobalKey<FormState>();
33   final firstNameController = TextEditingController();
34   final lastNameController = TextEditingController();
35   final midTermExamController = TextEditingController();
36   final finalTermExamController = TextEditingController();
37   final assignmentController = TextEditingController();
38   FieldOfStudy _selectedFieldOfStudy = FieldOfStudy.ine;
39   String? _selectedSubject;
40
41   @override
42   void dispose() {
43     firstNameController.dispose();
44     lastNameController.dispose();
45     lastNameController.dispose();
46     midTermExamController.dispose();
47     finalTermExamController.dispose();
48     assignmentController.dispose();
49     super.dispose();
50   }
51
52   InputDecoration _inputDecoration(
53     BuildContext context, {
54     required String hint,
55     required bool isFilled,
56   }) {
57     final errorColor = Theme.of(context).colorScheme.error;
58     final activeColor = isFilled ? Colors.green : Colors.grey;

```

```

59     const borderRadius = BorderRadius.all(Radius.circular(8.0));
60
61     return InputDecoration(
62       hintText: hint,
63       border: const OutlineInputBorder(borderRadius: borderRadius),
64       enabledBorder: OutlineInputBorder(
65         borderRadius: borderRadius,
66         borderSide: BorderSide(color: activeColor),
67       ),
68       focusedBorder: OutlineInputBorder(
69         borderRadius: borderRadius,
70         borderSide: BorderSide(color: activeColor, width: 4),
71       ),
72       errorBorder: OutlineInputBorder(
73         borderRadius: borderRadius,
74         borderSide: BorderSide(color: errorColor),
75       ),
76       focusedErrorBorder: OutlineInputBorder(
77         borderRadius: borderRadius,
78         borderSide: BorderSide(color: errorColor, width: 4),
79       ),
80       suffixIcon: isFilled
81         ? Icon(Icons.check_circle, color: Colors.green)
82         : null,
83     );
84   }
85
86   Widget _requiredTextField({
87     required BuildContext context,
88     required String label,
89     required String hint,
90     required String emptyMessage,
91     required TextEditingController controller,
92     required TextInputType keyboardType,
93   }) {
94     return Column(
95       crossAxisAlignment: CrossAxisAlignment.start,
96       children: [
97         Text(label, style: const TextStyle(fontSize: 18)),
98         const SizedBox(height: 8),
99         ValueListenableBuilder<TextEditingValue>(
100           valueListenable: controller,
101           builder: (context, value, _) {
102             final isFilled = value.text.trim().isNotEmpty;
103             return TextFormField(
104               controller: controller,
105               decoration: _inputDecoration(
106                 context,
107                 hint: hint,
108                 isFilled: isFilled,
109               ),
110               validator: (value) {
111                 if (value == null || value.trim().isEmpty) {
112                   return emptyMessage;
113                 }
114                 return null;
115               },
116               autovalidateMode: AutovalidateMode.onUnfocus,
117               keyboardType: keyboardType,
118               inputFormatters: keyboardType == TextInputType.number
119                 ? [FilteringTextInputFormatter.allow(RegExp(r'[0-9.]'))]
120                 : null,

```

```

121     );
122   },
123   ),
124   ],
125   );
126 }
127
128 @override
129 Widget build(BuildContext context) {
130   return Scaffold(
131     body: SafeArea(
132       child: SingleChildScrollView(
133         child: Padding(
134           padding: const EdgeInsets.symmetric(
135             vertical: 16.0,
136             horizontal: 24.0,
137           ),
138           child: Form(
139             key: _formKey,
140             child: Column(
141               crossAxisAlignment: CrossAxisAlignment.start,
142               children: [
143                 const Text(
144                   'Information Setup',
145                   style: TextStyle(
146                     fontSize: 18,
147                     fontWeight: FontWeight.bold,
148                     color: Color.fromARGB(255, 77, 77, 77),
149                   ),
150                 ),
151                 const Text('Please provide your basic personal information'),
152                 const SizedBox(height: 20),
153                 _requiredTextField(
154                   context: context,
155                   label: 'First Name *',
156                   hint: 'Jone',
157                   emptyMessage: 'Please enter your first name',
158                   controller: firstNameController,
159                   keyboardType: TextInputType.name,
160                 ),
161                 const SizedBox(height: 20),
162                 _requiredTextField(
163                   context: context,
164                   label: 'Last Name *',
165                   hint: 'Doe',
166                   emptyMessage: 'Please enter your last name',
167                   controller: lastNameController,
168                   keyboardType: TextInputType.name,
169                 ),
170                 const SizedBox(height: 20),
171                 const Text(
172                   'Field of Study *',
173                   style: TextStyle(fontSize: 18),
174                 ),
175                 const SizedBox(height: 8),
176                 FieldOfStudyRadio(
177                   value: _selectedFieldOfStudy,
178                   onChanged: (value) {
179                     setState(() => _selectedFieldOfStudy = value);
180                   },
181                 ),
182                 const SizedBox(height: 20),

```

43

```

183     const Text('Subject *', style: TextStyle(fontSize: 18)),
184     const SizedBox(height: 8),
185     DropdownButtonFormField<String>(
186       decoration: _inputDecoration(
187         context,
188         hint: 'Select Subject',
189         isFilled: false,
190       ),
191       items: const [
192         DropdownMenuItem(
193           value: 'mobile_application_development',
194           child: Text('MOBILE APPLICATION DEVELOPMENT'),
195         ),
196         DropdownMenuItem(
197           value: 'cloud_infra_software_&_service',
198           child: Text('CLOUD INFRA SOFTWARE & SERVICE'),
199         ),
200         DropdownMenuItem(
201           value: 'computer_programming',
202           child: Text('COMPUTER PROGRAMMING'),
203         ),
204         DropdownMenuItem(
205           value: 'computer_arch_&_operating_sys',
206           child: Text('COMP ARCH & OPERATING SYS'),
207         ),
208       ],
209       onChanged: (value) {
210         setState(() {
211           _selectedSubject = value;
212         });
213       },
214       validator: (value) {
215         if (value == null || value.isEmpty) {
216           return 'Please select a subject';
217         }
218         return null;
219       },
220     ),
221     const SizedBox(height: 20),
222     _requiredTextField(
223       context: context,
224       label: 'Midterm Exam *',
225       hint: '85',
226       emptyMessage: 'Please enter your midterm exam score',
227       controller: midTermExamController,
228       keyboardType: TextInputType.number,
229     ),
230     const SizedBox(height: 20),
231     _requiredTextField(
232       context: context,
233       label: 'Final Exam *',
234       hint: '90',
235       emptyMessage: 'Please enter your final exam score',
236       controller: finalTermExamController,
237       keyboardType: TextInputType.number,
238     ),
239     const SizedBox(height: 20),
240     _requiredTextField(
241       context: context,
242       label: 'Assignment *',
243       hint: '80',
244       emptyMessage: 'Please enter your assignment score',

```

44

```

245 controller: assignmentController,
246 keyboardType: TextInputType.number,
247 ),
248 const SizedBox(height: 30),
249 Row(
250   children: [
251     Expanded(
252       child: ElevatedButton(
253         style: ElevatedButton.styleFrom(
254           padding: const EdgeInsets.symmetric(vertical: 16),
255           shape: RoundedRectangleBorder(
256             borderRadius: BorderRadius.circular(8.0),
257           ),
258           backgroundColor: Colors.purple,
259         ),
260         onPressed: () {
261           if (!_formKey.currentState!.validate()) {
262             return;
263           }
264
265           Navigator.push(
266             context,
267             MaterialPageRoute(
268               builder: (context) => ResultPage(
269                 firstName: firstNameController.text.trim(),
270                 lastName: lastNameController.text.trim(),
271                 fieldOfStudy: _selectedFieldOfStudy.name,
272                 subject: _selectedSubject!,
273                 midTermScore: double.parse(
274                   midTermExamController.text.trim(),
275                 ),
276                 finalTermScore: double.parse(
277                   finalTermExamController.text.trim(),
278                 ),
279                 assignmentScore: double.parse(
280                   assignmentController.text.trim(),
281                 ),
282               ),
283             ),
284           );
285         },
286         child: const Text(
287           'Submit',
288           style: TextStyle(
289             color: Colors.white,
290             fontSize: 18,
291             fontWeight: FontWeight.bold,
292           ),
293         ),
294       ),
295     ],
296   ),
297 ],
298 ],
299 ),
300 ),
301 ),
302 ),
303 ),
304 );
305 }
306 }

```

45

```

307 // field_study_radio_btn.dart
308 import 'package:flutter/material.dart';
309
310 enum FieldOfStudy { ine, inet }
311
312 class FieldOfStudyRadio extends StatelessWidget {
313   const FieldOfStudyRadio({
314     super.key,
315     required this.value,
316     required this.onChanged,
317   });
318
319   final FieldOfStudy value;
320   final ValueChanged<FieldOfStudy> onChanged;
321
322   @override
323   Widget build(BuildContext context) {
324     final dividerColor = Theme.of(context).dividerColor;
325
326     Widget option({required String text, required FieldOfStudy value}) {
327       return InkWell(
328         borderRadius: BorderRadius.circular(8),
329         onTap: () => onChanged(value),
330         child: Padding(
331           padding: const EdgeInsets.symmetric(horizontal: 12, vertical: 10),
332           child: Row(
333             mainAxisAlignment: MainAxisAlignment.center,
334             children: [
335               Radio<FieldOfStudy>(
336                 value: value,
337                 visualDensity: VisualDensity.compact,
338               ),
339               Text(text),
340             ],
341           ),
342         ),
343       );
344     }
345
346     return RadioGroup<FieldOfStudy>(
347       groupValue: value,
348       onChanged: (value) {
349         if (value == null) return;
350         onChanged(value);
351       },
352       child: Container(
353         decoration: BoxDecoration(
354           border: Border.all(color: dividerColor),
355           borderRadius: BorderRadius.circular(8),
356         ),
357         child: Row(
358           children: [
359             Expanded(
360               child: option(text: 'INE', value: FieldOfStudy.ine),
361             ),
362             Container(width: 1, height: 44, color: dividerColor),
363             Expanded(
364               child: option(text: 'INET', value: FieldOfStudy.inet),
365             ),
366           ],
367         ),
368       ),

```

46

```

369     ),
370   );
371 }
372 }
373
374 // result.dart
375 import 'package:flutter/material.dart';
376
377 class ResultPage extends StatelessWidget {
378   const ResultPage({
379     super.key,
380     required this.firstName,
381     required this.lastName,
382     required this.fieldOfStudy,
383     required this.subject,
384     required this.midTermScore,
385     required this.finalTermScore,
386     required this.assignmentScore,
387   });
388
389   final String firstName;
390   final String lastName;
391   final String fieldOfStudy;
392   final String subject;
393   final double midTermScore;
394   final double finalTermScore;
395   final double assignmentScore;
396
397   double get totalScore {
398     return midTermScore + finalTermScore + assignmentScore;
399   }
400
401   String get grade {
402     final score = totalScore;
403     if (score >= 80) {
404       return 'A';
405     } else if (score >= 75) {
406       return 'B+';
407     } else if (score >= 70) {
408       return 'B';
409     } else if (score >= 65) {
410       return 'C+';
411     } else if (score >= 60) {
412       return 'C';
413     } else if (score >= 55) {
414       return 'D+';
415     } else if (score >= 50) {
416       return 'D';
417     } else {
418       return 'F';
419     }
420   }
421
422   String get formatSubject {
423     return subject.replaceAll('_', ' ').toUpperCase();
424   }
425
426   String get formatFieldOfStudy {
427     return 'Information and Network Engineering (${fieldOfStudy.toUpperCase()})';
428   }
429
430   String _formatScore(double value) {

```

47

```

431   if (value == value.roundToDouble()) {
432     return value.toStringAsFixed(0);
433   }
434   return value.toStringAsFixed(1);
435 }
436
437 Color _gradeColor(BuildContext context) {
438   final colorScheme = Theme.of(context).colorScheme;
439   final score = totalScore;
440   if (score >= 80) return Colors.green.shade700;
441   if (score >= 70) return Colors.teal.shade700;
442   if (score >= 60) return Colors.orange.shade700;
443   if (score >= 50) return Colors.deepOrange.shade700;
444   return colorScheme.error;
445 }
446
447 Widget _infoRow({
448   required IconData icon,
449   required String label,
450   required String value,
451 }) {
452   return Row(
453     crossAxisAlignment: CrossAxisAlignment.start,
454     children: [
455       Icon(icon, size: 18, color: const Color.fromARGB(255, 77, 77, 77)),
456       const SizedBox(width: 10),
457       Expanded(
458         child: Text('$label: $value', style: const TextStyle(fontSize: 15)),
459       ),
460     ],
461   );
462 }
463
464 Widget _card({required Widget child}) {
465   return Container(
466     width: double.infinity,
467     padding: const EdgeInsets.all(16),
468     decoration: BoxDecoration(
469       color: Colors.white,
470       borderRadius: BorderRadius.circular(12),
471       border: Border.all(color: const Color(0xFFE9E2F5)),
472       boxShadow: const [
473         BoxShadow(
474           color: Color(0x14000000),
475           blurRadius: 10,
476           offset: Offset(0, 6),
477         ),
478       ],
479     ),
480     child: child,
481   );
482 }
483
484 @override
485 Widget build(BuildContext context) {
486   final gradeColor = _gradeColor(context);
487
488   return Scaffold(
489     backgroundColor: const Color(0xFFFF7F3FF),
490     bottomNavigationBar: SafeArea(
491       minimum: const EdgeInsets.fromLTRB(24, 0, 24, 16),
492       child: SizedBox(

```

48

```

493 height: 52,
494 child: ElevatedButton(
495   style: ElevatedButton.styleFrom(
496     shape: RoundedRectangleBorder(
497       borderRadius: BorderRadius.circular(10),
498     ),
499     backgroundColor: Colors.purple,
500     foregroundColor: Colors.white,
501   ),
502   onPressed: () => Navigator.pop(context),
503   child: const Text(
504     'Back to Form',
505     style: TextStyle(fontSize: 16, fontWeight: FontWeight.bold),
506   ),
507 ),
508 ),
509 ),
510 body: SafeArea(
511   child: SingleChildScrollView(
512     padding: const EdgeInsets.symmetric(vertical: 16.0, horizontal: 24.0),
513     child: Column(
514       crossAxisAlignment: CrossAxisAlignment.start,
515       children: [
516         Text(
517           formatSubject,
518           style: const TextStyle(
519             fontSize: 20,
520             fontWeight: FontWeight.bold,
521             color: Color.fromARGB(255, 77, 77, 77),
522           ),
523         ),
524         const SizedBox(height: 4),
525         Text(
526           formatFieldOfStudy,
527           style: TextStyle(color: Colors.grey.shade700, height: 1.2),
528         ),
529         const SizedBox(height: 16),
530         _card(
531           child: Row(
532             children: [
533               Container(
534                 width: 52,
535                 height: 52,
536                 decoration: BoxDecoration(
537                   color: gradeColor.withValues(alpha: 0.12),
538                   shape: BoxShape.circle,
539                 ),
540                 alignment: Alignment.center,
541                 child: Text(
542                   grade,
543                   style: TextStyle(
544                     fontSize: 20,
545                     fontWeight: FontWeight.w800,
546                     color: gradeColor,
547                   ),
548                 ),
549               ),
550               const SizedBox(width: 14),
551               Expanded(
552                 child: Column(
553                   crossAxisAlignment: CrossAxisAlignment.start,
554                   children: [

```

49

```

555 Text(
556   '$firstName $lastName',
557   style: const TextStyle(
558     fontSize: 18,
559     fontWeight: FontWeight.w700,
560   ),
561 ),
562 const SizedBox(height: 4),
563 Text(
564   'Total Score: ${_formatScore(totalScore)}',
565   style: TextStyle(
566     fontSize: 14,
567     color: Colors.grey.shade700,
568   ),
569 ),
570 ],
571 ),
572 ),
573 ],
574 ),
575 ),
576 const SizedBox(height: 12),
577 _card(
578   child: Column(
579     crossAxisAlignment: CrossAxisAlignment.start,
580     children: [
581       const Text(
582         'Score Details',
583         style: TextStyle(
584           fontSize: 16,
585           fontWeight: FontWeight.w700,
586         ),
587       ),
588       const SizedBox(height: 12),
589       _infoRow(
590         icon: Icons.badge_outlined,
591         label: 'First Name',
592         value: firstName,
593       ),
594       const SizedBox(height: 10),
595       _infoRow(
596         icon: Icons.badge_outlined,
597         label: 'Last Name',
598         value: lastName,
599       ),
600       const SizedBox(height: 10),
601       const Divider(height: 20),
602       _infoRow(
603         icon: Icons.edit_note_outlined,
604         label: 'Mid-term',
605         value: _formatScore(midTermScore),
606       ),
607       const SizedBox(height: 10),
608       _infoRow(
609         icon: Icons.edit_note_outlined,
610         label: 'Final-term',
611         value: _formatScore(finalTermScore),
612       ),
613       const SizedBox(height: 10),
614       _infoRow(
615         icon: Icons.assignment_outlined,
616         label: 'Assignment',

```

50

```

617         value: _formatScore(assignmentScore),
618     ),
619     const SizedBox(height: 10),
620     const Divider(height: 20),
621     _infoRow(
622       icon: Icons.summarize_outlined,
623       label: 'Total',
624       value: _formatScore(totalScore),
625     ),
626     const SizedBox(height: 10),
627     _infoRow(
628       icon: Icons.emoji_events_outlined,
629       label: 'Grade',
630       value: grade,
631     ),
632   ],
633 ),
634 ),
635 ],
636 ),
637 ),
638 ),
639 );
640 }
641 }

```